

Haotian Xu 许皓天

RL algorithms and RL infrastructure for long-horizon agents — sole owner of the RL framework, RL training, and credit-assignment algorithms behind Apodex 1.0 and Apodex 1.1.

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2,000+ citations · h-index 16 · i10 22

SUMMARY

RL researcher and infrastructure engineer working on reinforcement learning for agents that run for thousands of steps. I built the asynchronous training stack for both Apodex flagship releases end to end, designed the fragment-level and token-level credit-assignment algorithms (PIVOT-RL, SAO), co-authored BPO, and ran the RL that produced the 1.0 → 1.1 benchmark jump — on a 397B-class MoE flagship and a 35B-A3B open-weight Mini. Co-author of OpenRLHF and 10+ papers at NeurIPS, ICLR, ACL, EMNLP, and TPAMI.

RESULTS APODEX 1.0 → 1.1

The RL stage I owned is what separates the two releases. Against Apodex 1.0, the 1.1 flagship gains **+35.0 on FrontierScience-Research** (28.3 → 63.3), **+22.0 on APEX-Agents** (16.5 → 38.5, 2.3×), **+19.5 on GDPVal** (59.3 → 78.8) and **+14.0 on FrontierFinance** (40.3 → 54.3), with simulated-business net worth on YC-Bench rising **21.6×** (\$47,966 → \$1,038,255); HLE (49.0 → 56.1) and DeepSearchQA (84.6 → 92.4) moved up too, confirming the agentic gains cost nothing in general reasoning or search. The 35B-A3B open-weight Mini reproduces the pattern on the same recipe at a twelfth of the scale (FrontierScience-Research +26.7, FrontierFinance +17.0, APEX-Agents +12.3), which is the evidence that the recipe rather than the parameter count is carrying the result. Artificial Analysis independently ranks Apodex 1.1 at **Intelligence Index 44, #11 of 177 models** against a median of 18, with GDPval-AA v2 Elo 1348 — ahead of DeepSeek V4 Pro and Qwen3.7 Max.

CORE TECHNICAL CONTRIBUTIONS

Fully-asynchronous agentic RL framework

Designed and built the training framework for both releases on top of miles, against the defining pathology of agentic RL: trajectory wall-clock cost is wildly irregular across search, code execution, file processing, and multi-agent coordination, so any synchronous pipeline idles on its slowest episode.

- **Fully decoupled loops** — training workers and rollout engines run on independent loops, communicating only through a shared data buffer and an output queue. Completed trajectories enter optimization asynchronously, with no barrier on stragglers.
- **Non-blocking evaluation pool** — a dedicated pool scores checkpoints without ever interrupting the training loop, taking evaluation off the critical path.
- **Hybrid-colocate GPU reclamation** — idle training cards are recruited as generation engines, converting bubble time into rollout throughput.

BPO: off-policy correction without importance sampling

RLVR is off-policy by construction: sync mini-batches, lagging async workers, partial rollouts spanning model versions, and train/inference numerical drift all separate the optimized policy from the sampling one. The standard fix collapses on long responses — at T = 16,384

with a per-token ratio of 1.001 the sequence IS weight is 1.29×10^7 against a linearized 17.4, so a handful of trajectories own the update. BPO drops importance sampling entirely.

- **Critic-free reconstruction of PMD** — policy mirror descent gives a per-token optimality condition on the rollout-policy advantage A^μ ; the Bellman equation under μ telescopes those advantages along the trajectory into $R(x,y) - V^\mu(x)$. Minimizing the squared trajectory residual under μ needs no importance weights and no separate critic, and provably shares PMD's optimal solution.
- **Gradient-equivalent to score centering** — linearizing BPO around $\pi = \mu$ yields the token gradient $-\hat{A} \cdot [g_t(y_t) - E_{v \sim \mu} g_t(v)]$, exactly what score centering constructs directly. The two coincide under the same $\hat{A}$, the same fixed μ , and exact full-vocabulary reverse KL — but they are different derivations (PMD optimality vs. removing score drift) and differ in how they approximate $\nabla \text{KL}(\mu \parallel \pi)$.
- **The approximation is the real difference** — BPO collapses the vocabulary to sampled-token-vs-rest (binary KL, smoothed ratio $\omega_t = (1+\epsilon-q_t)/(1+\epsilon-p_t)$, capped and sign-masked); score centering keeps the sampler's top-k ($k = 128$) and models the tail. BPO therefore stores one behavior probability per token ($O(1)$) and drops into an existing GRPO/PPO loss as a scalar weight, while score centering needs top-k IDs and probabilities per position ($O(k)$) plus a weighted score sum in the backward pass. BPO's correction stays collinear with the sampled token's score; score centering's does not.
- **Results** — Qwen3-30B-A3B-Base on DAPO-Math-17k with the policy loss as the only variable: peak three-benchmark average **50.5 vs CISPO 47.4, DPPO 46.4, GSPO 43.5, GRPO-ClipHigher 39.5** (AIME24/25/26 = 57.4 / 41.0 / 53.0) — **+3.1 over the strongest baseline, +11.0 over GRPO-ClipHigher**.

Write-ups — [BPO, part 1](#) · [BPO, part 2](#)

PIVOT-RL: tree-policy fragment selection

The headline RL algorithm of Apodex 1.1. A long agentic rollout terminates in a single 0/1 verifier reward: terminal outcomes give only coarse supervision, useful intermediate work routinely precedes a late failure, and successful traces still contain weakly-grounded decisions. PIVOT-RL makes the **trajectory fragment** the unit of credit — and picks which fragment to train on by tree search rather than by a heuristic cut.

- **Tree-policy pivot selection** — expand an on-policy search tree from the model's own rollout instead of cutting one linear trajectory after the fact. Sampling continuations from each candidate fork gives Q-value estimates at the branch, so the pivot is chosen on measured branch value. This costs more rollout compute than a heuristic cut and buys a markedly more reliable pivot-turn choice on long-horizon tasks.
- **Branch-point detection** — candidate forks come from first-error-step detection, high-entropy decision points, and offline LLM-judge screening for bad patterns, targeting the places where the policy starts following an unproductive strategy, leans on insufficient evidence, misuses a tool, or fails to revise an assumption.
- **Scaling** — on three held-out agentic evaluations (search, knowledge, science), scores improve consistently from 20 to 140 rollout steps.

Write-up — [Credit assignment after the terminal reward](#)

Critic pretraining: HL-Gauss loss and reasoning-trace warmup

A randomly-initialized value head makes every early advantage noise, and published single-rollout async PPO trains its critic with plain MSE. In miles-values I replaced that regression objective with classification and warm-started the critic on reasoning traces.

- **HL-Gauss value loss** — the value head emits num_bins logits over a categorical support; the scalar target is projected onto a Gaussian-smoothed distribution ($\sigma = 0.75 \times \text{bin width}$, ~6 bins of supervision) and trained with cross-entropy. Shipped at 51 bins over [0, 1]. Bounded softmax gradients tolerate noisy targets, and GAE is unaffected — it still reads a scalar $V = \text{softmax}(\text{logits}) \cdot \text{support}$.
- **Reasoning-trace warmup** — pretrain the critic on trajectories from earlier RL runs, so it enters PPO calibrated instead of climbing out of deeply negative explained variance.
- **IID pretrain/RL data beats more data** — with the critic corpus overlapping the RL set, EV holds ≈ 0.2 throughout and both AIME24/25 gain further; on a disjoint corpus the advantage decays back to the un-pretrained curve. Cheapest recipe: roll the critic corpus from the same SFT-filtered data used for RL.
- **Math results** — Qwen3.5-4B / deepmath-103k: HL-Gauss > MSE (MSE's AIME24 flat after 200 steps). Qwen3.5-35B-A3B / dap0-17k: all variants reach AIME24 0.91 / AIME25 0.85, but sample efficiency orders hl-gauss-pretrain > hl-gauss > MSE.
- **Agentic results** — 128k search-agentic task, single-rollout PPO on a deliberately small critic warmup (60–70k traces, 10 steps). By step 129 against baseline: **BrowseComp 58.0 → 70.0 (+12.0)**, **FrontierScience 45.0 → 53.3 (+8.3)**, **HLE 38.0 → 46.4 (+8.4)**, with APEX-Agents held flat (43.2 → 42.9).

Write-ups — [SAO, part 1](#) · [SAO, part 2](#) · [value pretraining](#)

Diagnostics for multi-week RL runs

Owned the instrumentation that kept 397B RL runs honest: loss-mask-aware explained variance aggregated across microbatch/DP/CP (the metric separating a learning critic from a silently dead one), rollout-structure collapse detection (parallel multi-query degradation under penalty-only rewards with length-normalized objectives), reward-hacking audits, benchmark-leakage closure for SWE-bench Verified-class evaluations, and FP8-vs-BF16 inference parity checks.

EXPERIENCE

Apodex *RL Algorithms & RL-Infra, Agentic RL* 2025.11 — Present

Formerly MiroMind. Sole owner of the RL framework, RL training, and credit-assignment algorithms for Apodex 1.0 and Apodex 1.1. Built the fully-async training stack, PIVOT-RL and SAO credit assignment; drove the 1.0 → 1.1 gains above on both the 397B flagship and the 35B-A3B open-weight Mini.

Xiaohongshu — HiLab *Math Reasoning & Agentic RL* 2024 — 2025.8

RL for mathematical reasoning and tool-using agents; agentic RL scaling laws for spontaneous code execution (NeurIPS 2025).

Douyin / ByteDance *AIGC Safety* 2023

Alibaba *Content Safety, Public Opinion Analysis, LLM Safety* 2018 — 2023

OPEN SOURCE

MILES-VALUES Dedicated SAO (Async Single-rollout PPO) framework with large-scale value pretraining.

ASYNC_OPENRLHF Fully-asynchronous RLHF pipeline extending OpenRLHF.

SELECTED PUBLICATIONS

Bellman Policy Optimization	arXiv:2609.15987 · 2026
Apodex 1.1: Scaling Agentic Intelligence for Complex Work	arXiv:2608.23283 · 2026
Apodex-1.0: A Verification-Centric Agent Team for Discoverative Intelligence	Apodex Tech Report · 2026
From System 1 to System 2: A Survey of Reasoning Large Language Models 529 cites	IEEE TPAMI 2025
OpenRLHF: A Ray-based Easy-to-use, Scalable and High-performance RLHF Framework 371 cites	EMNLP 2025
Agentic RL Scaling Law: Spontaneous Code Execution for Mathematical Problem Solving 71 cites	NeurIPS 2025
IterResearch: Rethinking Long-Horizon Agents with Interaction Scaling 44 cites	ICLR 2026
Search Self-Play: Pushing the Frontier of Agent Capability 25 cites	ICLR 2026
GEM: A Gym for Generalist LLMs 17 cites	ICLR 2026
Omni-DPO: A Dual-Perspective Paradigm for Dynamic Preference Learning 9 cites	ICLR 2026
TL;DR: Too Long, Do Re-weighting for Efficient LLM Reasoning Compression 10 cites	ACL 2026
Shuttle Between Instructions and Parameters	ACL 2026

EDUCATION

Tsinghua University <i>Department of Electronic Engineering</i>	2013 – 2016
UESTC <i>B.Eng.</i>	2009 – 2013

TECHNICAL PROFILE

RL ALGORITHMS	PPO/GRPO and group-normalized variants, GAE, value pretraining, off-policy importance correction and token-level clipping, fragment- and turn-level credit assignment, RLHF, MCTS, DPO variants, reward modeling and anti-reward-hacking.
RL INFRASTRUCTURE	Fully-async rollout/training decoupling, hybrid-colocate GPU reclamation, shared-buffer pipelines, Megatron and torch-dist checkpointing and conversion, TP/CP/EP parallelism, vLLM and SGLang rollout engines, FP8 inference parity, Ray, OpenRLHF, miles.
LANGUAGES	Chinese (native), English (professional).